

Combination

Random Variables

Plotting

- Plotting of the occurrences:
 - values on the **X-Axis**
 - Frequencies on the **Y-Axis**:
 - **Absolute frequencies**: # of occurrences
 - **Relative frequencies**: % of occurrences

Frequentist Approach

- In contrast to Bayesian Approach
 - Given: stable measurement conditions
 - Retrospectively determined **relative frequencies** of an occurrence
 - Serves: **Future Expectation** about the probability of an occurrence
- absolute frequencies $h(d_i)$
- relative frequencies: $f(d_i) = \frac{h(d_i)}{n}$

Definition RV

Formal Definition of a random variable Z :

$$Z : (\Omega, A, P) \rightarrow (\mathbb{R}, B, P_Z)$$

- mapping of true probability space (Triplet) to modelled probability space (Triplet)
 - Ω : set of base events - consists of discrete values that e_i can take
 - A : Event system
 - P : Probability measure - Probability for certain element of the event set to appear

Example: Flipping a coin

- Ω : Base events: $\{\text{head}, \text{tail}\} \cong \{0, 1\} \rightarrow \mathbb{R}$
- A : Event system: $B = \{\{\text{head}\}, \{\text{tail}\}, \{\text{head}, \text{tail}\}, \emptyset\}$
- P : Probability measure $P_Z(x) =$
 - 0.5 --- $x = 0$,
 - 0.5 ---- $x = 1$,
 - 0 --- else
- Application of Axioms of probability calculation rules
- e.g. $P(A \cup B) = P(A) + P(B)$

PDF

A 1-D function $f_Z(x)$ is a probability density function associated to the random variable Z iff

$$f_z(x) \geq 0 \text{ for } x \in \mathbb{R}$$

$f_z(x)$...is continuously defined for $x \in \mathbb{R}$

$$\int_{\mathbb{R}} f_Z(x) dx = \int_{-\infty}^{\infty} f_Z(x) dx = 1 = P(-\infty \leq Z \leq \infty)$$

Characteristic quantities of a pdf

So called **moment of a pdf** are used to characterise a pdf

k-th moment of a pdf:

$$M^k := \int_{-\infty}^{\infty} x^k f_Z(x) dx$$

First moment : expected value of a random variable

$$M^{(1)} := \int_{-\infty}^{\infty} x f_Z(x) dx := \mu_Z := E\{Z\}$$

Second moment:

$$M^{(2)} := \int_{-\infty}^{\infty} x^2 f_Z(x) dx := \mu_Z := E\{Z^2\}$$

Central Moment of a pdf

- *Characteristics of the measurement** can be described by the **characteristics of the pdf, i.e. the moments**
- **k-th central moment of a pdf, centered around expectation:**

$$D^{(k)} := \int_{-\infty}^{\infty} (x - \mu_Z)^k f_Z(x) dx = E\{(Z - E\{Z\})^k\}$$

2nd central moment: variance

$$D^{(2)} := \int_{-\infty}^{\infty} (x - \mu_Z)^2 f_Z(x) dx := \sigma_Z^2 = E\{(Z - E\{Z\})^2\} = \Sigma\{Z\}$$

- measure for variation around expected value
- spread of the data, statistical variance

k-th standardised moment of a pdf, centered and normalised:

$$\gamma^{(k)} := \frac{E\{(Z - E\{Z\})^k\}}{\sigma_Z^k}$$

skewness

$$\gamma^{(3)} := \frac{E\{(Z - E\{Z\})^3\}}{\sigma_Z^3}$$

- measure for the symmetry

kurtosis

$$\gamma^{(4)} := \frac{E\{(Z - E\{Z\})^4\}}{\sigma_Z^4}$$

- measure for the tail thickness

Multivariate PDF

A multivariate function $f_Z(x)$ is a multivariate probability density function associated to the random variable Z iff:

$$f_Z(x_1, x_2, \dots, x_n) = f_Z(x) \geq 0, \text{ for } x \in \mathbb{R}^n$$

$$f_Z(x) \dots \text{is continuously defined for } x \in \mathbb{R}^n$$

$$\int_{\mathbb{R}^n} f_Z(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_Z(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n = 1$$

Characteristics

All moments $M^{(k)}$, $D^{(k)}$ and $\gamma^{(k)}$ analogous to 1D pdf, now **vectors** in $\mathbb{R}^n$, important moments:

$$M^{(1)} := E\{Z\} = \begin{pmatrix} E\{Z_1\} \\ E\{Z_2\} \\ \dots \\ E\{Z_n\} \end{pmatrix}$$

with

$$E\{Z_i\} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} x_i f_Z(x_1, \dots, x_n) dx_1, \dots, dx_n$$

- mean value from an infinite number of realisations
- **probability-weighted average** of all possible values the random variable can take (Integration in case of real valued random variables)

2nd central moment: covariance

$$D^{(2)} := \Sigma\{Z\} = E\{(Z - E\{Z\})(Z - E\{Z\})^T\} = \begin{pmatrix} \sigma_{Z_1}^2 & \sigma_{Z_1 Z_2} & \dots & \sigma_{Z_1 Z_n} \\ \sigma_{Z_2 Z_1} & \sigma_{Z_2}^2 & \dots & \sigma_{Z_2 Z_n} \\ \dots & \dots & \dots & \dots \\ \sigma_{Z_n Z_1} & \sigma_{Z_n Z_2} & \dots & \sigma_{Z_n}^2 \end{pmatrix}$$

Covariance matrix with

$$\sigma_{Z_i}^2 = E\{(Z_i - E\{Z_i\})^2\} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_i - E\{Z_i\})^2 f_Z(x_1, \dots, x_n) dx_1 \dots dx_n$$

and

$$\sigma_{Z_i Z_j} = E\{(Z_i - E\{Z_i\})(Z_j - E\{Z_j\})\} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_i - E\{Z_i\})(x_j - E\{Z_j\}) f_Z(x_1, \dots, x_n) dx_1 \dots dx_n$$

- variation around the expected value
- relation to correlation: $\sigma_{Z_i Z_j} = \rho_{Z_i Z_j} \sigma_{Z_i} \sigma_{Z_j}$

Cumulative Distribution Function CDF F_Z

$$F_Z(c) := \int_{-\infty}^c f_Z(x) dx = P_{Z \leq c}$$

$$\frac{dF_Z}{dx} = f_Z(x)$$

Characteristics

- $F_Z(-\infty) = 0$
- $F_Z(\infty) = 1$
- $F_Z(c)$ is monotonically increasing

Probability for intervals:

$$P\{c_l \leq Z \leq c_u\} = F_Z(c_u) - F_Z(c_l)$$

Multivariate CDF

$$F : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$F_Z(c_1, c_2, \dots, c_n) = \int_{-\infty}^{c_1} \int_{-\infty}^{c_2} \dots \int_{-\infty}^{c_n} f_Z(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n$$

$$F_Z(c) = \int_{-\infty}^c f_Z(x) dx = P(Z \leq c) = P(Z_1 \leq c_1, Z_2 \leq c_2, \dots)$$

-> can be used to provide probability that a random variable takes a value from an interval: $P(c_l \leq Z \leq c_u) = F_Z(c_u) - F_Z(c_l)$

- monotonically non-decreasing function
- $0 \leq F_Z(c) \leq 1$ for any $c \in \mathbb{R}^n$
- $\lim_{c_i \rightarrow -\infty} F_Z(c_1, c_2, \dots, c_n) = 0$
- $\lim_{c_i \rightarrow \infty} F_Z(c_1, c_2, \dots, c_n) = 1$
- each random variable has a underlying pdf or distribution
- represents a kind of 'family' of all possible outcomes of the experiment -> not a specific result
- for a random variable we can provide probabilities, that the RV takes a value inside a specific interval
- $P_Z(Z \leq z) = F_Z(z)$
- $P_Z(z_l \leq Z \leq z_u) = F_Z(z_u) - F_Z(z_l)$

Uncorrelatedness vs. Independence

Given two random variables

$\mathcal{X}$: with pdf $f_{\mathcal{X}}$

and

$\mathcal{Z}$: with pdf $f_{\mathcal{Z}}$

X and Z are called uncorrelated iff $\Sigma\{X, Z\} = 0$

$$\rightarrow \Sigma\left\{\begin{bmatrix} \mathcal{X} \\ \mathcal{Z} \end{bmatrix}\right\} = \begin{bmatrix} \Sigma\{\mathcal{X}\} & 0 \\ 0 & \Sigma\{\mathcal{Z}\} \end{bmatrix}$$

$\mathcal{X}$ and $\mathcal{Z}$ are independent iff $P_{X,Z}(X \leq a, Z \leq b) = P_X(X \leq a)P_Z(Z \leq b)$

$$\rightarrow f_{X,Z}(u, v) = f_X(u)f_Z(v)$$

-> The pdf are factorisable. All joint higher moment vanish

- Independence is stronger, if X and Z are independent, they are uncorrelated (not the other way around)
- However in case X & Z are normally distributed, they are equivalent

Computational Rules

The Expectation operator $E\{\cdot\}$

$$E\{c\} = c$$

$$E\{a\mathbf{X}\} = aE\{\mathbf{X}\}$$

$$E\{\mathbf{X} + \mathbf{Y}\} = E\{\mathbf{X}\} + E\{\mathbf{Y}\}$$

$$E\{A\mathbf{X} + \mathbf{b}\} = AE\{\mathbf{X}\} + b$$

$$E\{E\{\mathbf{X}\}\} = E\{\mathbf{X}\}$$

Variance Propagation

Error has expected value 0

- $E\{\mathcal{E}\} = 0$
- other wise: systematic error in measurement

Covariance of Error

$$\Sigma\{\mathcal{L}\} = \Sigma\{\mathcal{E}\} := \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \dots & \sigma_{1n} \\ & \sigma_2^2 & \sigma_{23} & \dots & \sigma_{2n} \\ & & \dots & \dots & \dots \\ & & & \dots & \dots \\ & & & & \sigma_n^2 \end{bmatrix}$$

- σ_i^2 : variance of i th measurement (i th diagonal entry)
- σ_i : standard deviation of i th measurement (sqrt of i th diagonal entry)
- σ_{ij}^2 : covariance between i th and j th (dependency measure)
- p_{ij} : correlation $-1 \leq p_{ij} \leq 1$ (dependency measure), $p_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$

Accuracy & Propagation $\Sigma\{\mathcal{P}\}$ and $\Sigma\{\mathcal{A}\}$

General linear model:

$$y = \underset{[m \times 1]}{\phi} = \underset{[m \times n]}{F} \underset{[n \times 1]}{X} + \underset{[m \times 1]}{b}$$

We assume we are given:

$$\begin{aligned} E\{\mathcal{X}\} &= \mu_x = \dots \\ \Sigma\{\mathcal{X}\} &= \Sigma \end{aligned}$$

-> We want to determine $\mathcal{E}\{y\}$ and $\Sigma\{x\}$

$$y = FX + b$$

EV:

$$\begin{aligned} E\{y\} &= E\{FX + b\} \\ &= E\{FX\} + b \\ &= F\mu_x + b \\ &= \mu_y \end{aligned}$$

Variance:

$$\begin{aligned} \Sigma\{y\} &= \Sigma\{FX + b\} \\ &= E\{((FX + b) - E\{FX + b\})((FX + b) - E\{FX + b\})^T\} \\ &= E\{FX + b - FE\{X\} - b)(FX + b - FE\{X\} - b)^T\} \\ &= E\{(FX - FE\{X\})(FX - FE\{X\})^T\} \\ &= E\{(F(X - E\{X\}))(F(X - E\{X\}))^T\} \\ &= E\{F(X - E\{X\})(X - E\{X\})^T F^T\} \\ &= FE\{(X - E\{X\})(X - E\{X\})^T\} F^T \\ &= \Sigma\{X\} \\ &= F\Sigma\{X\}F^T \end{aligned}$$

... Law of variance (/error) propagation for linear transformation.

Remarks

- $\Sigma\{y\}^{-1}$ only exists if $\text{rank}(\mathbf{F}) = m < n$

In case

$$\mathcal{X} \sim \mathcal{N}(\mathbf{E}\{\mathcal{X}\}, \Sigma\{\mathcal{X}\})$$

$\mathcal{Y}$ follows a normal distribution as well

$$\mathcal{Y} \sim \mathcal{N}(\mathbf{F}\mathbf{E}\{\mathbf{X}\} + \mathbf{b}, \mathbf{F}\Sigma\{\mathbf{X}\}\mathbf{F}^T)$$

- This does not hold true for other distributions

$$\nabla\varphi(x) = J(\varphi(x)) = \begin{bmatrix} \nabla\varphi_1(x_1, \dots, x_n)^T \\ \vdots \\ \nabla\varphi_m(x_1, \dots, x_n)^T \end{bmatrix} = \begin{bmatrix} \frac{\partial\varphi_1(x)}{\partial x_1} & \dots & \frac{\partial\varphi_1(x)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial\varphi_m(x)}{\partial x_1} & \dots & \frac{\partial\varphi_m(x)}{\partial x_n} \end{bmatrix}$$

Expectation

$$\mathbf{E}\{\mathcal{Y}\} = \mathbf{E}\{\varphi(\mathcal{X})\} = \varphi(\mathbf{x})^T \big|_{\mathbf{x}=\mathbf{E}\{\mathcal{X}\}}$$

Variance

$$\Sigma\{\mathcal{Y}\} = \mathbf{F}\Sigma\{\mathcal{X}\}\mathbf{F}^T$$

with $\mathbf{F} = J(\varphi(x))$ - the Jacobian

- requires good approximation values to set up $\mathbf{F}$
 - we used $\mathbf{x}_0 = \mathbf{E}\{\mathcal{X}\}$

- Realisations of $\mathcal{X}$ to numerically set up $\mathbf{F}$

Least Squares Adjustment

- **Given:** Observations $\mathbf{l}$, the values of the matrix $\mathbf{A}$, and the weight matrix $\mathbf{P}$
- **Find:** Values $\tilde{\mathbf{x}}$ for unknown parameter $\mathbf{x}$ and values $\tilde{\mathbf{v}}$ for corrections $\mathbf{v}$ that
 - solve the function model (linear system)

$$\mathbf{l} + \mathbf{v} = \mathbf{Ax}$$

$$\begin{bmatrix} l_1 \\ \vdots \\ l_n \end{bmatrix} + \begin{bmatrix} n_1 \\ \vdots \\ n_n \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nm} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$$

- And minimise the objective: $\Omega(\mathbf{v}) = \mathbf{v}^T \mathbf{P} \mathbf{v}$

Solution:

$$\tilde{\mathbf{x}} = \mathbf{N}^{-1} \mathbf{n} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \mathbf{l}$$

$$\tilde{\mathbf{v}} = \mathbf{A} \tilde{\mathbf{x}} - \mathbf{l}$$

Control Computation

$$\mathbf{A}^T \mathbf{P} (\mathbf{A} \tilde{\mathbf{x}} - \mathbf{l}) = 0$$

$$\mathbf{A}^T \mathbf{P} \tilde{\mathbf{v}} = 0$$

Properties:

- $\mathbf{N} \tilde{\mathbf{x}} = \mathbf{n}$ is a linear system in $\tilde{\mathbf{x}}$ with m unknowns and m equations
- If $\text{rank}(\mathbf{a}) = m \Leftrightarrow \text{rank}(\mathbf{N}) = m \Rightarrow$ it has a unique solution
- $\mathbf{N}$ is a symmetric matrix as it is computed from the quadratic form $\mathbf{A}_{m \times n}^T \mathbf{P}_{n \times n} \mathbf{A}_{n \times m}$ with symmetric matrix $\mathbf{P}_{n \times n}$ as metric
- $\tilde{\mathbf{x}}$ is the best solution for the parameters for which $\mathbf{v}^T \mathbf{P} \mathbf{v}$ is minimal

Choosing a Weight Matrix

1. Inverted Covariance matrix

- $\mathbf{P} = \Sigma\{\mathbf{L}\}_{[n \times n]}^{-1}$
- In case of a diagonal matrix:

$$\blacksquare \mathbf{P} = \Sigma\{\mathbf{L}\}_{[n \times n]}^{-1} \begin{bmatrix} \frac{1}{\sigma_1^2} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\sigma_n^2} \end{bmatrix}$$

2. Separation into scalar σ_0^2 and Cofactor Matrix $\mathbf{Q}$

Non-Linear-Squares Adjustment

$$\begin{aligned}
l - l^0 + v &= \mathbf{A}_i \Delta x \\
\Delta l + v &= \mathbf{A} \Delta x \\
\begin{bmatrix} \Delta l_1 \\ \vdots \\ \Delta l_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} &= \begin{bmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nm} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \vdots \\ \Delta x_m \end{bmatrix}
\end{aligned}$$

1. Linearisation at x^0
 1. Compute Jacobian $A = \nabla \varphi(x) |_{x=x^0}$
 2. Compute Observations $l^0 = \varphi(x^0)$
 3. Reduce Observations $\Delta l = l - l^0$
2. Normal Equation: $N = A^T P A$ and $n = A^T P \Delta l$
3. Solution for parameter update: $\Delta \tilde{x} = N^{-1} n$
4. Solution for parameter: $\tilde{x} = x^0 + \Delta \tilde{x}$
5. Solution for Corrections: $\tilde{v} = A \Delta \tilde{x} - \Delta l$
6. Iterate with $x^0 = \tilde{x}$ until: $\varphi(\tilde{x}) - A \tilde{x} < \varepsilon$

Details: L9

LSA - Stochastic Approach

BLUE - Best Linear Unbiased Estimator

- **Linear** dependence on observations
 - $\tilde{\mathbf{X}} = \tilde{\mathbf{B}} \mathcal{L}$
- **Unbiased**: Expectation of estimator are true parameters
 - $E\{\tilde{\mathcal{X}}\} = \xi$
 - measurement error has expected value zero -> LS estimator is unbiased estimator
- **Best**: Smallest possible Covariance
 - $\Sigma\{\tilde{\mathcal{X}}\} \dots$ Minimal

LSA is BLUE

Cofactor, Covariance, Weight Matrix

$$\Sigma\{\mathcal{L}\} = \Sigma = \sigma_0^2 Q\{\mathcal{L}\} = \sigma_0^2 Q$$

$$P := Q^{-1}$$

$$\Sigma = \sigma_0^2 Q = \sigma_0^2 P^{-1} \quad \text{Covariance Matrix}$$

$$P = Q^{-1} = \frac{1}{\sigma_0^2} \Sigma^{-1} \quad \text{Weight Matrix}$$

$$Q = P^{-1} = \frac{1}{\sigma_0^2} \Sigma \quad \text{Cofactor Matrix}$$

With Variance Factor σ_0^2

- Estimated Parameters **are independent** of the variance factor ($\tilde{l}, \tilde{v}$)

- Covariance of estimated parameters **are not independent** of the variance factor

Helpful to:

- Estimation formulas can be simplified
- Unknown Variances-> Estimate optimal parameters w.r.t. unknown covariance of the observation

Estimation of Variance Factor

$$\tilde{\mathcal{S}}_0^2 = \frac{\tilde{\mathbf{v}}^T \mathbf{P} \tilde{\mathbf{v}}}{m - n} = \frac{\tilde{\mathbf{v}}^T \mathbf{Q}^{-1} \tilde{\mathbf{v}}}{r}$$

- Unbiased: $E\{\tilde{\mathcal{S}}_0^2\} = \sigma_0^2$
- Estimation of variance from residuals

Confidence Regions

$$P(\tilde{\mathbf{x}} - \sigma_{\tilde{\mathbf{x}}} k_{1-\alpha/2}^{\mathcal{N}(0,1)}) \leq \xi \leq \tilde{\mathbf{x}} + \sigma_{\tilde{\mathbf{x}}} k_{1-\alpha/2}^{\mathcal{N}(0,1)}) = \beta = 1 - \alpha$$

$$P(\xi - \sigma_{\tilde{\mathbf{x}}} k_{1-\alpha/2}^{\mathcal{N}(0,1)}) \leq \tilde{\mathbf{x}} \leq \xi + \sigma_{\tilde{\mathbf{x}}} k_{1-\alpha/2}^{\mathcal{N}(0,1)}) = \beta = 1 - \alpha$$

- Provide **Interval** where the true value is located in with probability β
- Accounting for error in estimate (measured by variance $\sigma_{\tilde{\mathbf{x}}}^2$)
- $\beta = 1 \Rightarrow k \rightarrow \infty$, but $\beta = 0.99 \Rightarrow k = 2.58$
- $\mathcal{N}(0, 1)$ to easily derive quantiles values $k_{1-\alpha/2}^{\mathcal{N}(0,1)}$

Hypothesis Testing

- Binary answer on verifying Null Hypothesis H_0 or alternative Hypothesis H_A
- Usually comparison between:
 - Realisations $\tilde{\mathbf{x}}$ of a random variable $\mathcal{X}$
 - or linear functional $\mathbf{B}^T \tilde{\mathbf{x}}$ of the random variable with a fixed constant value

Six-Pack for Hypothesis Testing

1. Definition of null and alternative hypothesis
 - Formalise H_0, H_A
2. Identify a test statistics & its distribution under assumption that H_0 is true
 - $\mathcal{T} := \dots \sim$ some normalised distribution
3. Define sensitivity of the test
 - Choose $\gamma = 1 - \alpha$... confidence level of a test, α ... significance level
4. Compute the value of the realisation of the test statistics from realisations
 - $t := \dots$ some specific number computed from a realisation
5. Compute the acceptance region A and rejection region R
 - Follow from critical values of the distribution of $\mathcal{T}$ and probability γ
6. Decision

- if $t \in A$: Accept H_0 , reject H_A or if $t \in R \notin A$: reject H_0 , accept H_A

Parameter Test

Assume that from a set of observations

$$\mathcal{L} \sim \mathcal{N}(\mathbf{A}\xi, \Sigma\{\mathcal{L}\}), \quad \text{with } \Sigma\{\mathcal{L}\} = \Sigma\{\mathcal{E}\} = \Sigma$$

we estimated some parameters in a least squares adjustment, then

$$\tilde{\mathcal{X}} \sim \mathcal{N}(\xi, \Sigma\{\tilde{\mathcal{X}}\})$$

From a specific measurements $\mathbf{l}$ (realization of $\mathcal{L}$) we estimated parameters

$$\tilde{\mathbf{x}} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \mathbf{l} \quad \text{with} \quad \Sigma\{\tilde{\mathcal{X}}\} = \sigma_0^2 (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} = \sigma_0^2 \mathbf{N}^{-1}$$

Note: here we assume σ_0^2 to be known ($\Sigma\{\tilde{\mathcal{X}}\}$ includes σ_0^2 and not $\tilde{\mathcal{S}}_0^2$).

Here we just consider a single parameter

$$\tilde{\mathcal{X}} = \tilde{\mathcal{X}}_i \quad \text{with} \quad \Sigma\{\tilde{\mathcal{X}}\} = \Sigma\{\tilde{\mathcal{X}}\}_{ii} = \sigma_{\tilde{\mathcal{X}}_i}^2 := \sigma_{\tilde{\mathcal{X}}}^2$$

$$\tilde{\mathcal{X}} \sim \mathcal{N}(E\{\mathcal{X}\}, \sigma_{\tilde{\mathcal{X}}}^2)$$

$$H_0 : E\{\tilde{X}\} = c \quad \text{vs.} \quad H_A : E\{\tilde{X}\} \neq c \quad \mathcal{T} = \frac{\tilde{X} - c}{\sigma_{\tilde{X}}^2} \sim \mathcal{N}(0, 1)$$

- ▶ Same answer as: Does our realization $\tilde{x}$ falls into the $1 - \alpha$ confidence region around c ? Yes $\rightarrow$ accept H_0 else reject H_0 and accept H_A
- ▶ The test statistic is only valid if $\sigma_{\tilde{X}}^2$ contains σ_0^2 , in case $\tilde{S}_0^2$ was used distribution changes from $\mathcal{N}(0, 1)$ to Student's t -distribution
- ▶ In case we accept H_A : decision is wrong with a probability $\alpha \rightarrow$ error of kind I; *producer risk*: we throw away out produced screws, because we tested them to have wrong length (although they haven't)
- ▶ In case we accept H_0 : results is wrong with a probability $\beta \rightarrow$ error of kind II; *consumer risk*: although we produced the screws with wrong length ($\neq c$) our test told us to accept H_0 , we sold incorrect products
- ▶ common values for α are 5 % or 1 %, if we lower α , β increases as the interval is larger



The Parameter Test



$$\tilde{X} \sim \mathcal{N}(c, \sigma_{\tilde{X}}^2)$$

Is there a significant difference between realization $\tilde{x}$ of $\tilde{X}$ and constant c ?

1. Hypothesis $H_0 : E\{\tilde{X}\} = c$ vs. $H_A : E\{\tilde{X}\} \neq c$

2. Test statistic

$$\tilde{\tau} = \frac{\tilde{x} - c}{\sigma_{\tilde{X}}} \sim \mathcal{N}(0, 1)$$

3. Sensitivity $\alpha =$

11

4. Realization of $\mathcal{T}$

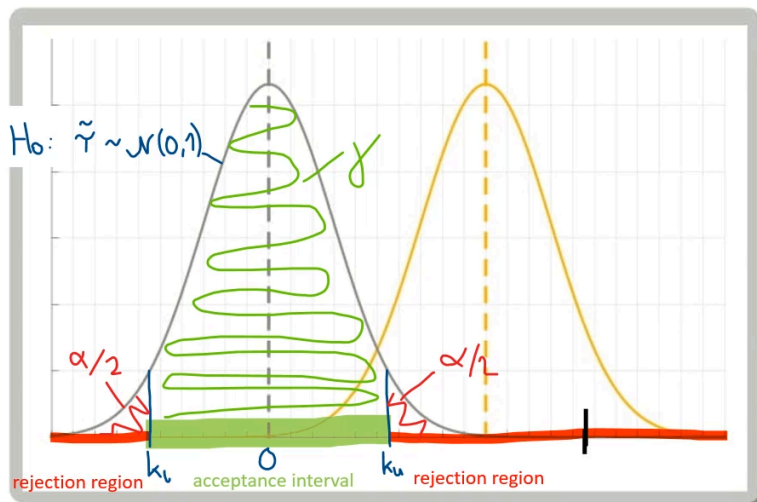
$$\tilde{t} = \frac{\tilde{x} - c}{\sigma_{\tilde{X}}}$$

5. Critical region: quantiles

$$k_v = -k_{1-\alpha/2}^{\mathcal{N}(0,1)} \quad k_u = k_{1-\alpha/2}^{\mathcal{N}(0,1)}$$

6. Decision

if $k_v \leq \tilde{t} \leq k_u$: we can not reject H_0
else : we reject H_0 and accept H_A



Global Model Test

Assume that from a set of observations

$$\mathcal{L} \sim \mathcal{N}(\mathbf{A}\xi, \Sigma\{\mathcal{L}\}), \quad \text{with } \Sigma\{\mathcal{L}\} = \Sigma\{\mathcal{E}\} = \Sigma = \sigma_0^2 \mathbf{Q} = \sigma_0^2 \mathbf{P}^{-1}$$

we estimated some parameters in a least squares adjustment, then

$$\tilde{\mathcal{X}} \sim \mathcal{N}(\xi, \Sigma\{\tilde{\mathcal{X}}\})$$

From a specific measurements $\mathbf{l}$ (realization of $\mathcal{L}$) we estimated parameters

$$\tilde{\mathbf{x}} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \mathbf{l} \quad \text{with} \quad \Sigma\{\tilde{\mathcal{X}}\} = \sigma_0^2 (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} = \sigma_0^2 \mathbf{N}^{-1}$$

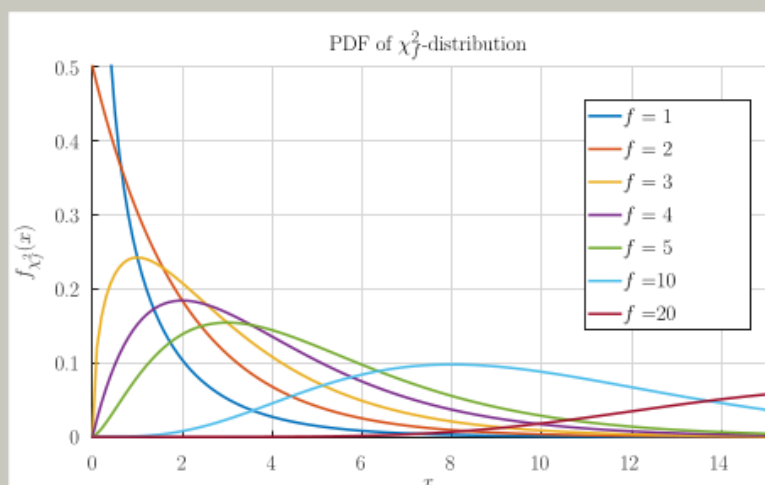
We can estimate the variance of unit weight

$$\tilde{s}_0^2 = \frac{\tilde{\mathbf{v}}^T \mathbf{P} \tilde{\mathbf{v}}}{n - m} \quad \text{or} \quad \tilde{s}_0^2 = \frac{\tilde{\mathbf{v}}^T \mathbf{P} \tilde{\mathbf{v}}}{n - m}$$

We want to check whether $\tilde{s}_0^2$ agrees with σ_0^2 accounting for the fact that $\tilde{s}_0^2$ is an estimated from a limited amount of data (and a single realization of $\tilde{\mathbf{v}}$)
 $\Rightarrow$ test of variances

The sum of f squared standard normally (i.e. $\mathcal{N}(0, 1)$) distributed random variables follows has a χ_f^2 distribution with f degrees of freedom.

PDF of the χ_f^2 Distribution depending on f



$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \mathbf{x} \in \mathbb{R}^{f \times 1}$$

$$\begin{aligned} y &:= \mathbf{x}^T \mathbf{x} = \sum_{i=1}^f x_i^2 \\ &= x_1^2 + x_2^2 + \dots + x_f^2 \\ y &\sim \chi_f^2 \end{aligned}$$

Moments of χ_f^2 Distribution

$$\begin{aligned} E\{\chi_f^2\} &= f \\ \Sigma\{\chi_f^2\} &= 2f \end{aligned}$$

1. Hypothesis

$$H_0 : E\{\tilde{S}_0^2\} = \sigma_0^2 \quad \text{vs.} \quad H_A : E\{\tilde{S}_0^2\} > \sigma_0^2$$



2. Test statistic

$$\begin{aligned} \mathcal{T} &= \tilde{\mathbf{v}}^T \Sigma^{-1} \tilde{\mathbf{v}} = \frac{\tilde{\mathbf{v}}^T \mathbf{P} \tilde{\mathbf{v}}}{\sigma_0^2} \\ &= (n-m) \frac{\tilde{S}_0^2}{\sigma_0^2} \sim \chi_{n-m}^2 \end{aligned}$$

3. Sensitivity

$$\alpha = \text{some } \%$$

4. Realization of $\mathcal{T}$

$$\tilde{t} = (n-m) \frac{\tilde{\mathbf{v}}^T \mathbf{P} \tilde{\mathbf{v}}}{\sigma_0^2} = (n-m) \frac{s_0^2}{\sigma_0^2}$$

5. Critical region: quantile

$$k_\alpha = \text{chi2inv}(n-m)$$

6. Decision

We can use the test of variances to validate our adjustment model.

$$H_0 : E\{\tilde{S}_0^2\} = \sigma_0^2 \quad \text{vs.} \quad H_A : E\{\tilde{S}_0^2\} > \sigma_0^2 \quad \mathcal{T} = \frac{\tilde{\mathbf{v}}^T \mathbf{P} \tilde{\mathbf{v}}}{\sigma_0^2} \sim \chi_{n-m}^2$$

In case H_A accepted:

With chosen significance level of α , the estimate $\tilde{s}_0^2$ is larger than σ_0^2

- ▶ given σ_0^2 is wrong, observations are less accurate than assumed $\leadsto$ continue analysis with $\tilde{s}_0^2$
- ▶ or our functional model is wrong, e.g. observations contain more signal than modeled ($\tilde{\mathbf{v}}$ too large)
- ▶ or there are outliers in the data (large entries in $\tilde{\mathbf{v}}$)

The test can not determine what is wrong (stochastic/functional model, data). With probability α , decision can be wrong as well.

Helpers

From Rad to Deg:

$$\text{Deg} = \frac{180^\circ}{\pi} \times \text{Rad}$$

$$\text{Rad} = \frac{\pi}{180^\circ} \times \text{Rad}$$

Common Position-Related Functions and Their Derivatives

1. Euclidean Distance ($\mathbb{R}^2$)

Function:

$$d(x, y) = \sqrt{(x - x_0)^2 + (y - y_0)^2}$$

Partial derivatives:

$$\frac{\partial d}{\partial x} = \frac{x - x_0}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}, \quad \frac{\partial d}{\partial y} = \frac{y - y_0}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}$$

2. Squared Euclidean Distance

Function:

$$d^2 = (x - x_0)^2 + (y - y_0)^2$$

Derivatives:

$$\frac{\partial d^2}{\partial x} = 2(x - x_0), \quad \frac{\partial d^2}{\partial y} = 2(y - y_0)$$

3. Polar Coordinates

Transformation:

$$x = r \cos \theta, \quad y = r \sin \theta$$

Derivatives:

w.r.t. r :

$$\frac{\partial x}{\partial r} = \cos \theta, \quad \frac{\partial y}{\partial r} = \sin \theta$$

w.r.t. θ :

$$\frac{\partial x}{\partial \theta} = -r \sin \theta, \quad \frac{\partial y}{\partial \theta} = r \cos \theta$$

4. Circular Motion (Trigonometric Position)

Functions:

$$x(t) = R \cos(\omega t), \quad y(t) = R \sin(\omega t)$$

Velocity (1st derivatives):

$$\dot{x}(t) = -R\omega \sin(\omega t), \quad \dot{y}(t) = R\omega \cos(\omega t)$$

Acceleration (2nd derivatives):

$$\ddot{x}(t) = -R\omega^2 \cos(\omega t), \quad \ddot{y}(t) = -R\omega^2 \sin(\omega t)$$

5. Arc Length

Function:

$$s(t) = \int \|\mathbf{r}'(t)\| dt$$

Derivative:

$$\frac{ds}{dt} = \|\mathbf{r}'(t)\|$$

6. Angle from Coordinates (atan2)

Function:

$$\theta = \text{atan2}(y, x)$$

Derivatives:

$$\frac{\partial \theta}{\partial x} = -\frac{y}{x^2 + y^2}, \quad \frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2}$$

7. Unit Direction Vector

Function:

$$\mathbf{u} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$$

Derivative (Jacobian):

$$\frac{\partial \mathbf{u}}{\partial \mathbf{r}} = \frac{1}{\|\mathbf{r}\|} (I - \mathbf{u}\mathbf{u}^T)$$

8. Distance to a Line

Function:

$$d = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$$

Derivatives (ignoring absolute value sign):

$$\frac{\partial d}{\partial x} = \frac{a}{\sqrt{a^2 + b^2}}, \quad \frac{\partial d}{\partial y} = \frac{b}{\sqrt{a^2 + b^2}}$$

9. Gaussian Function (Localization)

Function:

$$f(x) = \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Derivative:

$$f'(x) = f(x) \cdot \left(-\frac{x-\mu}{\sigma^2}\right)$$

10. Exponential Decay

Function:

$$f(d) = e^{-\lambda d}$$

Derivative:

$$f'(d) = -\lambda e^{-\lambda d}$$

11. Logistic Function

Function:

$$f(x) = \frac{1}{1 + e^{-x}}$$

Derivative:

$$f'(x) = f(x)(1 - f(x))$$

12. Linear Motion

Function:

$$x(t) = x_0 + vt$$

Derivative:

$$\frac{dx}{dt} = v$$

Basic Rules for Constructing Derivatives

1. Constant Rule

Function:

$$f(x) = c$$

Derivative:

$$f'(x) = 0$$

2. Power Rule

Function:

$$f(x) = x^n$$

Derivative:

$$f'(x) = nx^{n-1}$$

3. Constant Multiple Rule

Function:

$$f(x) = c \cdot g(x)$$

Derivative:

$$f'(x) = c \cdot g'(x)$$

4. Sum Rule

Function:

$$f(x) = g(x) + h(x)$$

Derivative:

$$f'(x) = g'(x) + h'(x)$$

5. Difference Rule

Function:

$$f(x) = g(x) - h(x)$$

Derivative:

$$f'(x) = g'(x) - h'(x)$$

6. Product Rule

Function:

$$f(x) = g(x) \cdot h(x)$$

Derivative:

$$f'(x) = g'(x)h(x) + g(x)h'(x)$$

7. Quotient Rule

Function:

$$f(x) = \frac{g(x)}{h(x)}$$

Derivative:

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2}$$

8. Chain Rule

Function:

$$f(x) = g(h(x))$$

Derivative:

$$f'(x) = g'(h(x)) \cdot h'(x)$$

9. Exponential Function

Function:

$$f(x) = e^x$$

Derivative:

$$f'(x) = e^x$$

10. General Exponential Function

Function:

$$f(x) = a^x$$

Derivative:

$$f'(x) = a^x \ln(a)$$

11. Natural Logarithm

Function:

$$f(x) = \ln(x)$$

Derivative:

$$f'(x) = \frac{1}{x}$$

12. General Logarithm

Function:

$$f(x) = \log_a(x)$$

Derivative:

$$f'(x) = \frac{1}{x \ln(a)}$$

13. Trigonometric Functions

Sine:

$$\frac{d}{dx}(\sin x) = \cos x$$

Cosine:

$$\frac{d}{dx}(\cos x) = -\sin x$$

Tangent:

$$\frac{d}{dx}(\tan x) = \frac{1}{\cos^2 x}$$

14. Inverse Trigonometric Functions

Arcsine:

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

Arccosine:

$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

Arctangent:

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$